When Do Kolmogorov–Arnold Networks Help in Wireless? A Matched-Capacity, Non-Saturated Benchmark

Phuc Hao Do[a]

[a]*Danang Architecture University, 566 Nui Thanh, Da Nang, Vietnam*

Abstract

Kolmogorov–Arnold networks (KANs) move the nonlinearity of a multilayer perceptron (MLP) from nodes onto learnable univariate edge functions, and are widely reported to match or beat MLPs at fewer parameters while staying interpretable. We ask *when* an edge-wise basis helps, and *why*. Using a parameter-matched, multi-seed protocol with paired confidence intervals, we evaluate 4 KAN instantiations against MLP, convolutional and classical baselines on three non-saturated signal-processing regimes: power-amplifier modeling, OFDM channel estimation, and modulation classification on RML2016.10a. We then isolate the governing variable. On a synthetic family whose difficulty a neutral control holds fixed to within 0.065 dB while cross-coordinate interaction varies, the KAN advantage decays monotonically with interaction: at 256 training samples it reverses sign, MLPs winning by 1.97 to 3.24 dB on purely interacting targets, whereas at 4096 samples it erodes from 10.18 to 4.75 dB without reversing. All instantiations order the same way; the reversal survives matching network depth on both sides, and at the reversal point sweeping 8 KAN configurations still leaves the matched MLP ahead by 1.7 dB. Given a shared convolutional backbone, an MLP head reaches 0.6893 against 0.6755 for the best KAN head, so the classification gap is not an inductive-bias artifact, and the usual convolutional baseline is itself under-tuned by 0.0324. Symbolic extraction, the step that would make a KAN readable, costs over a decibel. We map two axes of where edge-wise bases help, and release the protocol and harness. Code and data: `https://github.com/haodpsut/kan-wireless-benchmark`.

Keywords: Kolmogorov–Arnold networks, learning systems, benchmarking, parameter-matched comparison, interpretability, signal processing

1. Introduction

Kolmogorov–Arnold networks (KANs) [1] replace the fixed node nonlinearities of a multilayer perceptron (MLP) with *learnable* univariate functions on the edges, parameterized by splines [1] or truncated Fourier series [2]. Two properties are repeatedly advertised: a favorable accuracy-per-parameter trade-off against MLPs, and interpretability, because the learned one-dimensional edge functions can in principle be read off and replaced by closed-form symbols. Within roughly a year these claims have propagated into wireless communications: KANs have been proposed for modulation recognition, channel estimation, digital predistortion (DPD) [3], radio-frequency fingerprinting [4], and 6G signal intelligence [5].

The wireless literature, however, has adopted KANs largely in a *showcase* mode: a KAN is shown to beat some baseline on some task, and the advantage is attributed to the architecture. Two methodological problems make such claims hard to trust, and both are known in the broader machine-learning community [6]. First, **capacity is rarely matched**: a small KAN is compared against a differently sized MLP, so "fewer parameters" can be an artifact of the comparison rather than a property of the model. Second, **benchmarks are often saturated**: when every model already exceeds, say, 99% accuracy, differences are within noise and the comparison cannot distinguish architectures at all. A recent survey of KANs [7] and a fairness study on vision and tabular tasks [6] both caution that KAN advantages shrink or vanish once these controls are applied, but the question has not been asked cleanly *for wireless*, where the tasks, inductive biases, and baselines differ.

This paper asks the question directly and answers it with measurement. We do not propose a new architecture; we propose an honest evaluation and report what it shows, whichever way it falls. Concretely:

- We define a **parameter-matched, multi-seed benchmarking protocol** (Section 3) in which KANs and baselines are sized to the same parameter budget, trained identically, and compared with bootstrap confidence intervals and paired differences. Baselines are chosen to be *strong and task-appropriate*, not strawmen.
- We evaluate three wireless regimes chosen to be **non-saturated**, so differences are resolvable: (i) power-amplifier behavioral modeling / DPD, a smooth low-dimensional regression that is the regime most favorable to KANs (Section 4); (ii) automatic modulation classification (AMC) on the real RML2016.10a dataset (Section 5); and (iii) OFDM channel estimation, an estimation task with a strong LMMSE baseline (Section 6).
- We conduct an **interpretability audit** (Section 7) that sep-

Email address: `haodp@dau.edu.vn` (Phuc Hao Do)

arates two things a claim of interpretability must satisfy: are the learned edges simple, and is the simple form free? We find the edges are simple but the symbolic distillation that realizes the interpretability claim degrades accuracy.

- We run **ablations** (Section 8) that stress-test the comparison in KAN's favor: parameter-budget, *symmetric* KAN-and-MLP hyperparameter, training-set-size, and SNR sweeps. These reveal honest crossovers, KANs win in some regimes, and pin down where.
- We audit **computational cost** (Section 9): at matched parameters, KANs are 4 to 30× slower per inference than the MLP, so the "efficiency" that motivates them is largely illusory once compute, not just parameter count, is measured.
- We synthesize a **regime map** and practical guidance (Section 11): KANs are competitive and win in specific regimes (high-SNR classification, low data, richer bases) but never dominate, are beaten by the right inductive bias, and carry a measurable interpretability cost. We release all code (Section 12).

Our aim is neither to promote nor to dismiss KANs, but to replace showcase claims with a regime map: to show precisely where the imported value propositions hold and where they do not, and to give the community a protocol that surfaces this before a claim is believed.

2. Background and Related Work

2.1. Kolmogorov–Arnold networks

KANs are motivated by the Kolmogorov–Arnold representation theorem [8], which states that any multivariate continuous function can be written as a finite composition of continuous univariate functions and addition. Where the classical universal-approximation results for MLPs [9, 10] place the nonlinearity at the nodes and learn linear edge weights, KANs [1, 11] invert this: the edges carry learnable univariate functions and the nodes merely sum. A KAN layer maps $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ to $\mathbf{y} \in \mathbb{R}^{d_{\text{out}}}$ through a learnable univariate function on each edge:

$$y_j = \sum_{i=1}^{d_{\text{in}}} \phi_{ji}(x_i) + b_j, \tag{1}$$

where each ϕ_{ji} is a spline [1] or a truncated Fourier series [2]. The spline variant we use follows the standard efficient formulation [2]: $\phi_{ji}(x) = w^b_{ji}\,\text{silu}(x) + \sum_c w^s_{jic} B_c(x)$, with B_c the B-spline basis on a fixed grid. The Fourier variant uses $\phi_{ji}(x) = \sum_{k=1}^{K} [a_{jik}\cos(kx) + b_{jik}\sin(kx)]$. Both give a per-layer parameter count of the form $d_{\text{in}} d_{\text{out}} C$, where C is the per-edge coefficient count; this is the quantity we hold fixed when we speak of a matched budget. A rapidly growing family of variants swaps the edge basis (wavelets in Wav-KAN [12], Gaussian radial bases in FastKAN [13], Chebyshev polynomials in Cheby-KAN [14]), and recent work also examines the hardware inference cost of these choices [15]; none changes the parameter accounting above. Our matched-budget comparison therefore covers the two most common instantiations on the three wireless regimes, and extends to 4 instantiations, adding FastKAN [13] and Cheby-KAN [14], in the controlled experiment of Section 10, spline and Fourier, is representative. We use PyTorch implementations [16] following the efficient spline formulation [2].

2.2. KAN claims and their audits

The original proposal [1] emphasizes accuracy-per-parameter on scientific regression and symbolic-regression interpretability. Subsequent scrutiny is mixed: [6] finds that under a fair comparison KANs beat MLPs mainly on symbolic formula tasks and often lose on vision and tabular data, and a broad survey [7] catalogues the fast-growing but largely showcase-style application literature. The interpretability claim in particular is usually asserted qualitatively (a plot of a few edges) rather than tested for fidelity. This matters because interpretability in machine learning is a substantive property, not a byproduct of a readable plot: the literature distinguishes genuinely interpretable models from post-hoc explanations of black boxes [17, 18, 19], and the question of whether a simpler surrogate preserves a model's behavior is exactly the subject of knowledge distillation and model-compression studies [20, 21]. Our fidelity audit (Section 7) applies that lens to KAN's symbolic-extraction claim.

2.3. KANs in wireless

Deep learning is now pervasive in the wireless physical layer [22, 23], and KAN adoption there is recent and application-driven. For DPD, several KAN and symbolic-KAN models have been proposed to linearize power amplifiers with reduced complexity [3, 24, 25]; for radio-frequency fingerprinting, KAN-augmented residual networks report identification gains [4]; and signal-intelligence surveys for 6G list KANs among emerging tools alongside large language models [5, 26]. Across these works, the comparison is typically against an unmatched MLP or a single baseline, and the benchmark is not checked for saturation. The tasks themselves are well studied independently: AMC on the RML2016 datasets [27, 28, 29] is a standard classification benchmark, best attacked with convolutional inductive bias [30], whose low-SNR band remains far from saturation [31, 32]; PA behavioral modeling has a mature classical baseline in the memory polynomial and its generalizations [33, 34, 35]; and richer real-world benchmarks such as DeepSense 6G [36] confirm that many wireless tasks are far from solved. Interpretable-AI work for the physical layer [37] motivates the interpretability question but does not quantify KAN fidelity. Our contribution is to place KANs on these tasks under a matched-capacity, non-saturated, multi-seed protocol and to audit interpretability quantitatively.

3. Benchmarking Protocol

The protocol (Fig. 1) exists to remove the two confounds identified in Section 1. It has four rules.

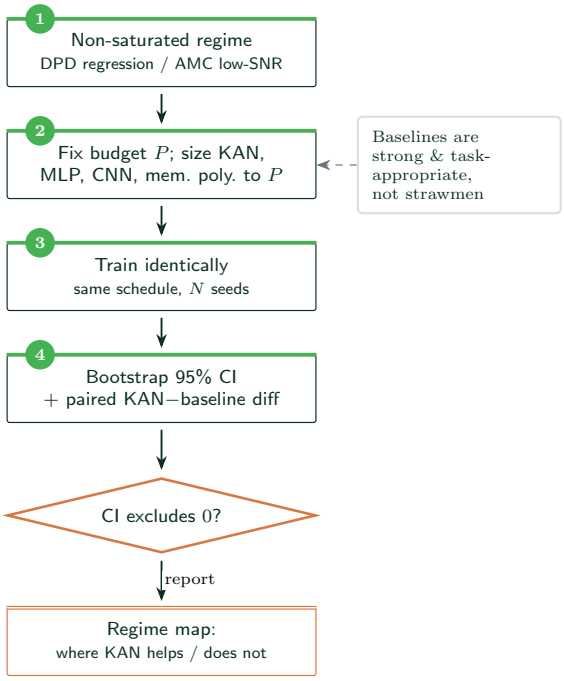


Figure 1: The honest benchmarking protocol: fix a parameter budget, size every model to it, train identically over multiple seeds, and report bootstrap intervals and paired differences on a non-saturated regime. Baselines are strong and task-appropriate.

(1) Matched capacity. For each task we fix a target parameter budget P and size every model to it. Write d for the input width, h for the hidden width and c for the output width. The MLP of the main tables has *two* hidden layers, $d \to h \to h \to c$ with ReLU at the nodes; each KAN has *one* hidden layer, $d \to h \to c$, with the learned univariate function on every edge. Their parameter counts are

$$P_{\mathrm{MLP}} = dh + h^2 + hc + (2h + c), \tag{2}$$

$$P^{\mathrm{F}}_{\mathrm{KAN}} = 2K\,(dh + hc) + (h + c), \tag{3}$$

$$P^{\mathrm{S}}_{\mathrm{KAN}} = (1{+}G{+}k)\,(dh + hc), \tag{4}$$

where K is the Fourier order and (G, k) the spline grid and degree; the h^2 term in (2) is the MLP's second hidden layer, and the spline layer of (4) carries no separate bias because its base branch supplies one. The basis factor $2K$ or $(1{+}G{+}k)$ multiplies the KAN's own $dh{+}hc$ connectivity, so a KAN and an MLP reach the same P only at different hidden widths.

Matching P at unmatched depth is a confound, and we treat it as one rather than leaving it implicit: Section 10.4 repeats the controlled experiment with depth matched on both sides and reports what changes and what does not. We fix the Fourier-KAN, compute its P from (3), and solve (2) and (4) for the MLP and spline-KAN widths that land within a few percent of it. This makes "KAN uses fewer parameters" testable rather than assumed. For DPD all learned models sit at $P \approx 2330$ parameters; the residual mismatch is under 3%.

(2) Strong, task-appropriate baselines. A KAN must be compared against the baseline a practitioner would actually use.

Table 1: Models and matched-capacity sizing. Each learned model is sized to the shared per-task budget P; the CNN is included as the strong structural baseline for AMC.

Model	Edge / layer function	Sized by
KAN (spline)	B-spline on each edge	hidden width
KAN (Fourier)	truncated Fourier per edge	width, order K
MLP	linear + ReLU nodes	hidden width
CNN	1-D conv. over I/Q	fixed (small)
mem. poly.	$x\lvert x\rvert^{k-1}$ basis, LS fit	memory, order

For the smooth regression regime that is a memory polynomial fit by regularized least squares [34], the classical DPD workhorse; we condition its basis so it is not a strawman. For the structured classification regime it is a small convolutional network that exploits the I/Q time structure. The matched MLP is the fairness control common to both.

(3) Multiple seeds and interval estimates. Each configuration is trained over several seeds (five for the regression regime, ten for classification); throughout, ± denotes one standard deviation across seeds, and bracketed intervals are bootstrap 95% confidence intervals of the mean. For the head-to-head question we report the *paired* KAN-minus-baseline difference with its own bootstrap interval, and call a difference significant only when that interval excludes zero. No result rests on a single seed.

(4) Non-saturated regimes. We deliberately choose operating points where no model is near the ceiling, so a real difference can appear. This is the opposite of the common practice of reporting on a band where every model exceeds 99%.

Both KAN variants, the MLP, the CNN, and the memory-polynomial baseline are implemented in a single harness so that training schedule, optimizer, and data pipeline are identical across models; only the architecture changes. All neural models are trained with the Adam optimizer [38] under a cosine learning-rate schedule and identical epoch budgets, and the memory-polynomial baseline is fit by regularized least squares. Table 1 summarizes the models and how each is sized to the shared budget.

4. Regime 1: PA Behavioral Modeling (DPD)

4.1. Task and data

Power-amplifier (PA) behavioral modeling is a smooth, low-dimensional nonlinear regression and is the regime KANs are *claimed* to handle best, since each edge can in principle capture a one-dimensional AM/AM-like curve. We drive a Wiener–Hammerstein PA with a pulse-shaped 16-QAM baseband signal at 6 dB input backoff. With input FIR h_1, output FIR h_2, and $u[n] = (h_1 * x)[n]$, the static Saleh nonlinearity [35] applies amplitude and phase distortion

$$A(r) = \frac{\alpha_a r}{1 + \beta_a r^2}, \qquad \Phi(r) = \frac{\alpha_p r^2}{1 + \beta_p r^2}, \tag{5}$$

so that the PA output is

$$y[n] = \big(h_2 * v\big)[n], \quad v[n] = A(|u[n]|)\, e^{\,j\left(\angle u[n] + \Phi(|u[n]|)\right)}. \tag{6}$$

Table 2: DPD / PA modeling, matched capacity, 10 seeds. NMSE in dB (lower is better).

Model	Params	NMSE [dB] ↓
MLP (matched)	2354	$\mathbf{-16.32 \pm 0.10}$
GMP (classical)	–	-16.20 ± 0.05
KAN, Fourier	2330	-15.51 ± 0.06
KAN, spline (canon.)	2280	-14.02 ± 0.09

Because (5)–(6) is not a memory polynomial, the classical memory-polynomial baseline gets no unfair "true model class" advantage; all learners approximate. The task is to predict the complex PA output $y[n]$ from a length-M=5 memory window of the complex input, and the metric throughout the paper is the normalized mean-squared error

$$\mathrm{NMSE} = 10 \log_{10} \frac{\sum_n \|\hat{y}[n] - y[n]\|^2}{\sum_n \|y[n]\|^2} \ [\mathrm{dB}]. \tag{7}$$

Models are trained over 5 seeds. We evaluate *behavioral modeling*, the forward-model accuracy that a predistorter is built on, rather than a full indirect-learning DPD loop; we therefore report NMSE and not ACPR or EVM, and read the results as a statement about learning a smooth PA nonlinearity, not about end-to-end linearization. Because the ground-truth PA is a Saleh model, its AM/AM curve is smooth by construction; a measured PA with strong memory or hysteresis could shift the boundaries, as we note in Section 11.

4.2. Results

Table 2 and Fig. 2 report the outcome. At a matched budget of 2330 parameters, the plain MLP achieves the best NMSE, −16.32 dB, beating the Fourier-KAN (−15.51 dB) and the canonical spline-KAN (−14.02 dB). The paired differences are +0.81 dB (Fourier-KAN worse than MLP) and +2.3 dB (spline-KAN worse than MLP); both confidence intervals exclude zero, so the MLP advantage is significant. Crucially, the classical baseline is not a strawman: a properly built generalized memory polynomial (GMP) [34],

$$\hat{y}[n] = \sum_{k \text{ odd}} \sum_{m} \sum_{|m'-m| \le 2} c_{k,m,m'} \, x[n-m] \left| x[n-m'] \right|^{k-1}, \tag{8}$$

with the lagging/leading cross-envelope terms $m' \neq m$ that a diagonal memory polynomial omits, reaches −16.2 dB, statistically tied with the best neural model and better than both KANs. So at the *default* matched configuration, even on the regime most favorable to KANs, the KANs beat neither a same-budget MLP nor the right classical method, and the widely used spline-KAN is the worst option of all. This ranking is not the whole story, however: Section 8 shows that a Fourier-KAN with a richer basis overtakes the MLP at matched capacity, and that KANs win when training data is scarce. The point of the default result is only that KAN superiority is not automatic; it must be earned and measured.

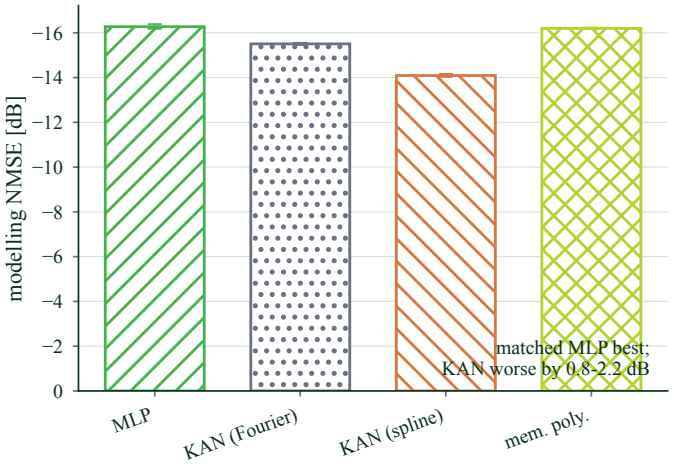


Figure 2: DPD regime, matched capacity. A same-budget MLP achieves the best modeling NMSE; both KAN variants are worse (paired differences 0.81–2.3 dB, intervals exclude zero).

5. Regime 2: Modulation Classification (RML2016.10a)

5.1. Task and data

Automatic modulation classification is a high-dimensional, structured-input classification task. We use the real RML2016.10a dataset [28], which contains 128-sample complex I/Q frames over 11 modulations and a range of SNRs. We restrict to the *low-SNR band* from −6 to 6 dB, where accuracy is far from saturation and model differences are resolvable, and draw 40000 power-normalized frames. The flat models see the 256-dimensional flattened I/Q vector; the CNN sees the 2×128 frame. Models are trained over 10 seeds.

5.2. Results

Table 3 and Fig. 3 report the outcome. The small convolutional baseline, with only 5291 parameters, reaches 0.658 accuracy and dominates every flat model. Among the flat models at a matched budget of about 68k parameters, the MLP (0.495) again beats both KANs; the spline-KAN (0.468) is close but below, and the Fourier-KAN (0.349) is markedly worse. The paired Fourier-KAN-minus-MLP difference is −0.146 (KAN worse), and the CNN leads the best KAN by +0.190; both intervals exclude zero. One lesson is unconditional and mirrors nothing in the regression regime: the right inductive bias, a convolution over the I/Q sequence, beats all flat models by a wide margin at a fraction of the parameters. The MLP-over-KAN ordering, however, is specific to this low-SNR band, and it inverts elsewhere: the SNR sweep in Section 8 shows the spline-KAN overtaking the MLP above about 2 dB and across the full SNR range, so our choice of the hardest band, made to avoid saturation, happens to be where the MLP looks best. This is exactly why an honest benchmark must resolve the operating point rather than report a single aggregate.

5.3. Where the KAN advantage lives

To locate the KAN advantage we resolve accuracy by modulation on the mid-band (0 to 12 dB), the regime where the SNR

Table 3: AMC on RML2016.10a, low-SNR band [−6, 6] dB, matched flat-model capacity, 10 seeds. Accuracy (higher is better).

Model	Params	Accuracy ↑
CNN (task-appropriate)	5291	**0.658 ± 0.005**
MLP (matched)	68651	0.495 ± 0.003
KAN, spline (canon.)	69420	0.468 ± 0.006
KAN, Fourier	68395	0.349 ± 0.007

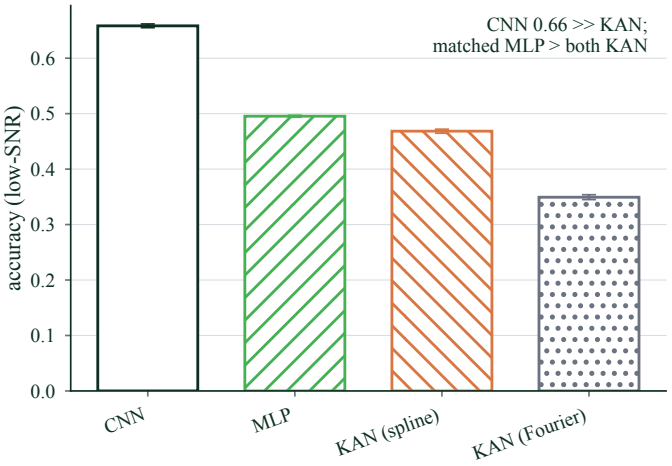


Figure 3: AMC regime (real RML2016.10a, low-SNR). A small CNN dominates; among matched flat models the MLP beats both KANs. The correct inductive bias, not the edge parameterization, decides the task.

sweep places the spline-KAN ahead of the MLP; this decomposes the high-SNR crossover, not the low-SNR headline of Table 3, and we report it as such. In this regime (Fig. 4) the spline-KAN's advantage over the matched MLP is not spread evenly: it concentrates on the phase- and amplitude-shift modulations whose decision regions are constellation geometry, gaining +0.23 on BPSK, +0.20 on PAM4, and +0.18 on 8PSK, while it slightly trails the MLP on the analog and frequency-modulated classes (for example −0.10 on GFSK). This pattern is consistent with, though it does not prove, a per-edge mechanism: a learnable univariate function per input can carve the smooth amplitude/phase boundaries of a constellation, but offers little for signals whose class is carried by frequency dynamics that a flat model reads only weakly in any case. The KAN is thus not a generically better classifier; where it helps, it is a better fit to a specific, identifiable structure.

5.4. Is the classification gap just the convolution?

A fair reading of Table 3 is that the CNN wins because it sees the I/Q time structure while every flat model sees a 256-dimensional vector, so the comparison may measure the value of convolution rather than the value of edge-wise nonlinearity. The objection is sound, and it has a direct test: hold the feature extractor fixed and vary only the head.

We take the convolutional stack of the same CNN, freeze the architecture, and attach four heads at matched head budgets: the original linear head, an MLP head, a Fourier-KAN head, and a spline-KAN head, each trained end to end over 10 seeds.

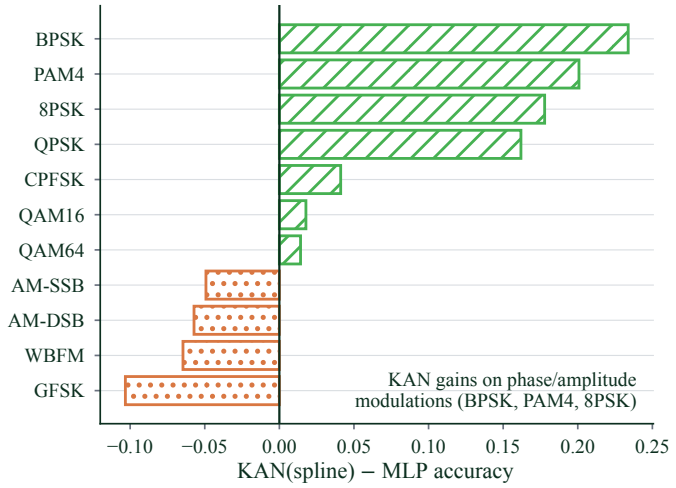


Figure 4: Per-modulation accuracy difference, spline-KAN minus matched MLP, on the mid-band (0 to 12 dB) where the KAN leads overall. The KAN advantage concentrates on constellation-geometry modulations (BPSK, PAM4, 8PSK, QPSK) and reverses on frequency/analog modulations, consistent with a per-edge decision-boundary mechanism.

The linear head reproduces the CNN of Table 3 at 0.6569. The MLP head reaches 0.6893 and the spline-KAN head 0.6755; the paired difference between them is −0.0138 in favour of the MLP head, with the interval excluding zero. The Fourier-KAN head does not train reliably in this setting at all, reaching 0.2112 with a seed-to-seed standard deviation of the same order as its mean, and we report that instability rather than omitting the configuration.

Two conclusions follow, and one of them is uncomfortable for us. First, the classification gap is not an artefact of convolutional inductive bias: given the same features, edge-wise nonlinearity does not beat node-wise nonlinearity here, so the objection is answered and the regime map stands. Second, our own "task-appropriate" baseline was under-tuned: simply replacing its linear head with an MLP head buys 0.0324 accuracy at the same backbone. A paper whose argument is that strong baselines are routinely under-tuned should say plainly when its own was, and the number above is the correction.

6. Regime 3: OFDM Channel Estimation

To test a third, structurally different task, we add OFDM channel estimation: an *estimation* problem rather than PA modeling or classification. On an N=64 subcarrier grid with L=8 multipath taps (exponential power-delay profile), comb pilots observe the channel frequency response at 16 subcarriers under noise at 10 dB SNR, and the task is to estimate the channel at all subcarriers. The metric is channel NMSE in dB, over 5 seeds. The decisive point is the baseline: linear minimum mean-square error (LMMSE) estimation is the Wiener estimator, optimal in the linear-Gaussian setting, and is the genuine classical workhorse here. From the noisy pilot observations $\mathbf{y}_p = \mathbf{H}_p + \mathbf{n}$ it forms

$$\hat{\mathbf{H}} = \mathbf{R}_{HH_p}(\mathbf{R}_{H_pH_p} + \sigma^2\mathbf{I})^{-1}\mathbf{y}_p, \tag{9}$$

Table 4: OFDM channel estimation at 10 dB SNR, matched capacity, 10 seeds. NMSE in dB (lower is better).

Model	NMSE [dB] ↓
LMMSE (classical)	**−13.36 ± 0.01**
MLP (matched)	−13.26 ± 0.02
KAN, spline (canon.)	−13.27 ± 0.01
KAN, Fourier	−13.11 ± 0.01
linear interpolation	−9.48 ± 0.01

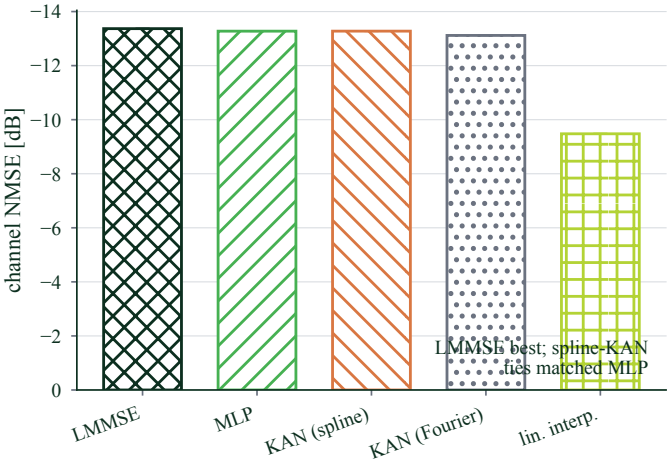


Figure 5: Channel estimation, matched capacity. Classical LMMSE is the best estimator; the spline-KAN exactly ties the matched MLP and neither beats LMMSE, mirroring the PA-modeling regime on a different task type.

with $\mathbf{R}_{HH_p}$ and $\mathbf{R}_{H_pH_p}$ the channel cross- and pilot auto-covariances estimated from training; we also include simple linear interpolation as a weak reference.

Table 4 and Fig. 5 report the result. The classical LMMSE is the best estimator at −13.36 dB, ahead of every neural model. Among the neural models the spline-KAN (−13.27 dB) exactly ties the matched MLP (−13.26 dB), the Fourier-KAN (−13.11 dB) is marginally behind, and all three clear the linear-interpolation reference (−9.48 dB) by a wide margin. As in the PA-modeling regime, the KAN buys nothing over a same-size MLP, and the right classical method is not beaten by either. Three regimes, three task types, and the same conclusion at the default configuration: KANs are competitive but not superior, and never beat the appropriate classical or structural baseline.

7. Interpretability Audit

The second KAN claim is interpretability: the learned univariate edges are simple enough to read off and replace by closed-form symbols, yielding an interpretable model "for free." A claim of this kind must satisfy two conditions, and we test them separately on the DPD KAN, where the task is smooth and thus most favorable to the claim.

(Q1) Simplicity. Are the learned first-layer edge functions well approximated by a human-readable low-order polynomial? We sample each edge on its input support and fit a cubic (over 5 seeds). The fits are excellent: the median per-edge R^2 is 0.998

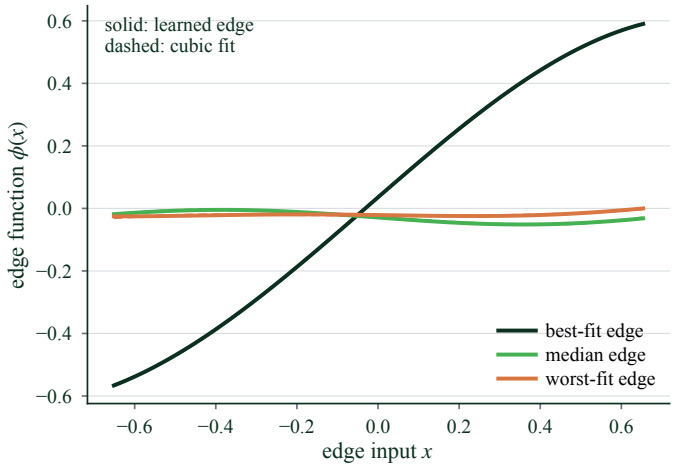


Figure 6: Three representative learned KAN edge functions (solid) and their cubic fits (dashed), spanning the best, median, and worst per-edge fit. The fits are nearly coincident, confirming the edges are simple, readable curves. Note the near-flat median and worst-fit edges carry little of the model, which is why the variance-weighted R^2 is reported.

and 99.7% of edges exceed $R^2 = 0.95$ (Fig. 7). To guard against near-flat edges inflating this, we also weight each edge's R^2 by its function variance; the variance-weighted mean is 0.998, so the high-amplitude edges that actually carry the model are simple too, not just the negligible ones. Fig. 6 makes this concrete by overlaying three representative learned edges, spanning the best, median, and worst fits, on their cubics: the dashed fits sit essentially on top of the solid curves, and the highest-magnitude edge is a simple monotonic S-shape reminiscent of an AM/AM characteristic. The edges are indeed simple; the claim is not vacuous.

(Q2) Fidelity: is it free? We then *distill* the first layer into those cubics (a symbolic surrogate), keep the rest of the network exact, and re-measure NMSE over the same 5 seeds. The surrogate degrades NMSE by 1.17 ± 0.21 dB. In other words, the readable form is not the performant form: extracting the interpretable symbols discards more than a decibel of modeling accuracy, consistently across seeds. Because we symbolize only the first layer, this is a *lower bound* on the cost of a fully symbolic KAN, and a task-loss-refit surrogate might recover part of it; even so, the naive extraction that the interpretability claim relies on is not free. This is precisely the nuance that qualitative "plot a few edges" demonstrations omit.

8. Ablations: Stress-Testing the Comparison

A single matched-budget comparison could be unfair to KANs in ways that a proponent would rightly object to: perhaps the budget is too small, the KAN is under-tuned, the data too plentiful, or the operating point unrepresentative. We therefore run four ablations, each designed to give the KAN its best shot, and report the honest crossovers they reveal.

Budget sweep (Fig. 8). We sweep the DPD parameter budget from about 0.8k to 9.0k. The MLP-over-KAN ordering persists at every budget, and the gap does not close: the Fourier-KAN

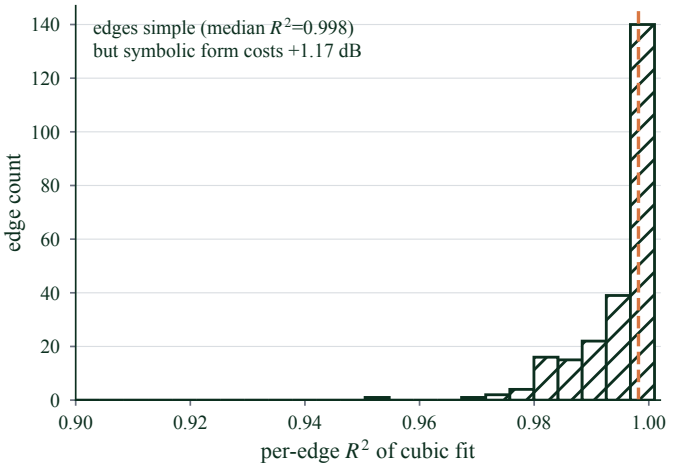


Figure 7: Interpretability audit on the DPD KAN. The learned edges are simple (a cubic fits with median $R^2 = 0.998$), but distilling them into that symbolic form costs 1.17 dB of NMSE: the interpretable model is not the performant one.

improves with budget but tracks below the MLP, and the spline-KAN plateaus near −14 dB and stops improving. Giving the KAN more parameters of the same kind does not rescue it here.

KAN and MLP hyperparameters (Tables 6 and 5). Fairness demands that if we tune the KAN, we tune the MLP too. We do both. Increasing the Fourier basis order to $K = 8$ raises the Fourier-KAN to −17.14 dB at 4634 parameters, and spectral richness, not width, is the lever. But the MLP is not fixed either: sweeping its depth and activation (Table 5), a three-layer MLP reaches −16.72 dB at the default budget, beating every matched-budget KAN there, and −16.95 dB at the larger $K = 8$ budget. So the Fourier-KAN's regression win survives a *symmetrically* tuned MLP, but thinly, and only at the larger budget.

How thinly is a question the first version of this comparison could not answer, because it reported two independently averaged means and no interval. We therefore repeated exactly that comparison as a *paired* test over 20 seeds, training both models on identical data draws. The paired difference is −0.156 dB in favour of the Fourier-KAN, with a bootstrap 95% interval of [−0.246, −0.066] dB and an exact Wilcoxon signed-rank $p = 0.007296$. The advantage is therefore real at this budget, and it is also smaller than the per-seed spread of either model taken alone: the standard deviation of the paired difference is larger than its mean, so the result is visible only because the comparison is paired. We state this explicitly because the unpaired version of the same numbers would not have supported a directional claim, and a reader who checks our earlier arithmetic rather than our protocol would have been right to object.

This is the flip side of the capacity-confound argument: just as an unmatched KAN can win for the wrong reason, a well-chosen KAN can win for the right one, and only a matched, symmetrically tuned, *paired* comparison tells them apart.

Sample efficiency (Fig. 9). A distinct claimed KAN strength is data efficiency. It holds here: with 1k and 4k training examples the Fourier-KAN beats the matched MLP (paired differences −0.23 and −0.37 dB, intervals excluding zero), and the ordering

Table 5: Symmetric MLP sweep on DPD at the default budget (5 seeds; the main regression tables use 10). A tuned three-layer MLP is the strongest matched-budget model, so the KAN is not beaten by an untuned baseline. The default 2-layer row (−16.25) is this sweep's own run; it matches Table 2's −16.32 within seed noise.

MLP configuration	Params	NMSE [dB] ↓
1 layer, ReLU	2329	−15.22 ± 0.14
2 layer, ReLU (default)	2354	−16.25 ± 0.08
2 layer, GELU	2354	−14.48 ± 0.22
2 layer, tanh	2354	−15.78 ± 0.13
3 layer, ReLU	2389	**−16.72 ± 0.17**
4 layer, ReLU	2277	−16.60 ± 0.10

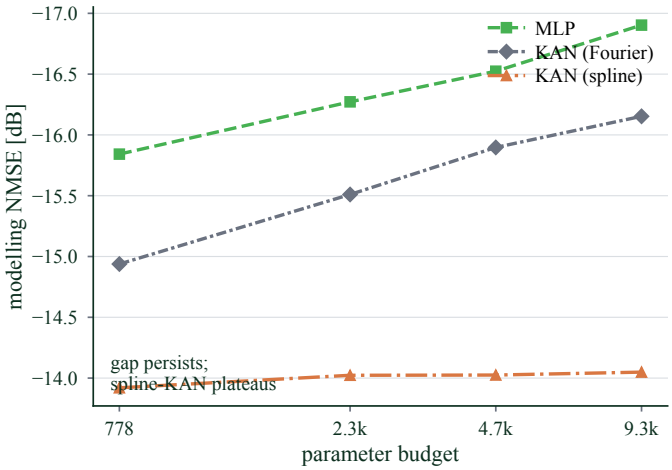


Figure 8: DPD budget sweep. The MLP leads at every budget; the Fourier-KAN improves but tracks below it, and the spline-KAN plateaus. More same-kind parameters do not close the gap.

inverts only once data is plentiful (+0.76 dB at the full set). A KAN is a reasonable choice when labeled data is the binding constraint.

SNR sweep (Fig. 10). Finally we resolve the AMC accuracy by SNR instead of aggregating over a band. The convolutional baseline leads at every SNR. Among the flat models, the MLP is best only at low SNR; the spline-KAN overtakes it above about 2 dB (reaching 0.620 versus the MLP's 0.480 at 10 dB), and over the *full* SNR range the spline-KAN is the best flat model (0.375 versus 0.333). The aggregate ranking is thus an artifact of which band one reports, which is the saturated-benchmark problem in a subtler form: not only can a band be too easy, it can be chosen (even inadvertently) to favor one model.

8.1. The symmetric sweep, extended to all three regimes

The sweep above tunes the MLP only on the regression task, which leaves the obvious objection that the other two regimes compare a tuned KAN against a default MLP. We therefore ran the same sweep, six depth-and-activation configurations at the matched budget over 10 seeds, on channel estimation and on modulation classification.

The result does not rescue the KANs anywhere, and in both regimes it widens the gap. On channel estimation the reference

Table 6: DPD KAN hyperparameter sweep (10 seeds). A richer Fourier basis (K=8, −17.14 dB at 4634 params) beats even a symmetrically tuned MLP (−16.95 dB at that budget), but by only −0.156 dB when the comparison is paired over 20 seeds (Section 8).

Variant	Setting	Params	NMSE [dB] ↓
Fourier	$K = 2$	1178	-14.42 ± 0.05
Fourier	$K = 4$	2330	-15.54 ± 0.09
Fourier	$K = 8$	4634	$\mathbf{-17.14 \pm 0.12}$
spline	grid 4	1152	-13.03 ± 0.15
spline	grid 6	1440	-14.04 ± 0.13
spline	grid 10	2016	-15.09 ± 0.09

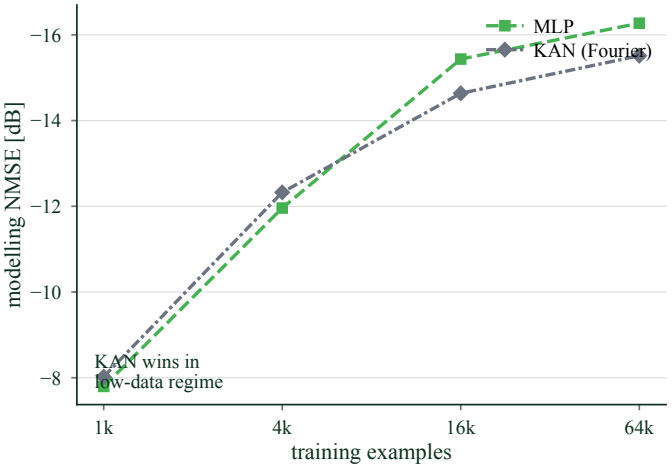


Figure 9: DPD sample efficiency. The Fourier-KAN beats the matched MLP when training data is scarce (≤ 4k examples); the ordering inverts only once data is plentiful.

Fourier-KAN reaches −13.114 dB while the best swept MLP reaches −13.28 dB; every one of the six configurations beats the KAN, with paired intervals excluding zero. On modulation classification the reference Fourier-KAN reaches 0.3517 accuracy against 0.501 for the best swept MLP, again with all six configurations ahead. The asymmetry objection is thus answered in the direction that is inconvenient for the architecture under test, which is the only direction in which answering it is informative.

9. Computational Cost: The Efficiency Illusion

Matching parameter count, the axis on which KAN "efficiency" is usually claimed, hides the axis that matters for deployment: compute. A KAN edge evaluates a spline or a truncated Fourier series where an MLP edge evaluates a single multiply-add, so at equal parameters a KAN does substantially more work. Counting multiply-adds per layer,

$$C_{\mathrm{MLP}} = 2\, d_{\mathrm{in}} d_{\mathrm{out}}, \tag{10}$$

$$C^{\mathrm{F}}_{\mathrm{KAN}} = 2K\, d_{\mathrm{in}} d_{\mathrm{out}} + \kappa\,(2K)\, d_{\mathrm{in}}, \tag{11}$$

$$C^{\mathrm{S}}_{\mathrm{KAN}} = (G{+}k)\, d_{\mathrm{in}} d_{\mathrm{out}} + \rho\, d_{\mathrm{in}}, \tag{12}$$

where κ is the cost of a cos / sin pair and ρ that of the B-spline recurrence. Two effects compound: the basis multiplies the connectivity term by $2K$ or $G{+}k$, and the transcendental or recurrence terms add fixed per-edge overhead that no parameter count reflects. The parameter budget absorbs the multiplicative factor by shrinking the hidden width, but the per-edge overhead does not shrink with it; it reappears as latency.

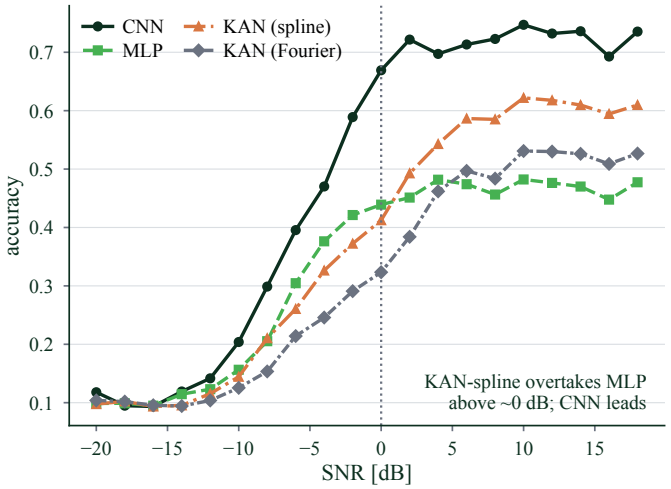


Figure 10: AMC accuracy resolved by SNR. The CNN leads throughout; among flat models the MLP wins only at low SNR, while the spline-KAN overtakes it above ∼ 2 dB. Reporting a single band hides this crossover.

Table 7: Compute at matched parameters. Inference time per batch (256) and slowdown versus the matched MLP. KANs cost far more compute for the same parameter count.

Task	Model	Params	Infer. [ms] ↓
DPD	MLP	2354	0.04
DPD	KAN, Fourier	2330	0.27
DPD	KAN, spline	2280	1.14
AMC	CNN	5291	25.2
AMC	MLP	68651	0.16
AMC	KAN, Fourier	68395	0.74
AMC	KAN, spline	69420	3.47

Table 7 measures it. At the *same* parameter budget, inference is 7× to 30× slower for the KANs than for the matched MLP on the regression models, and 4.5× to 21× slower on the classification models, with the canonical spline-KAN the most expensive by far. So even in the regimes where a KAN edges out the MLP on accuracy, it pays with an order of magnitude more compute per inference, and a practitioner optimizing performance per FLOP or per millisecond, the realistic constraint at a base station or on an edge device, would not choose it. "Fewer parameters" is the wrong metric; matched-parameter is already generous to the KAN. These timings are single-thread CPU measurements; on a massively parallel accelerator the wall-clock gap would narrow, since the extra per-edge operations parallelize. The *arithmetic*, however, does not: the multiply-add and transcendental counts (10)–(12) are device-independent, so the energy and throughput penalty remains even where latency is hidden. We report CPU latency as the transparent proxy and the FLOP counts as the device-independent statement.

10. Why the Map Looks Like This: An Additivity Experiment

Sections 4–6 say *where* an edge-wise basis helps; they do not say *why*, and a regime map without a mechanism is hard to transfer to a new task. This section supplies the missing variable and tests it directly.

10.1. Hypothesis

An MLP applies its nonlinearity at the *nodes*: every hidden unit sees a linear mixture of all inputs, so a univariate dependence on one coordinate must be synthesised from mixtures. A KAN applies its nonlinearity at the *edges*: each edge carries its own univariate function of a single coordinate. For d inputs and a basis of size G per edge, the first KAN layer therefore spends $O(dG)$ parameters on *strictly univariate* degrees of freedom. The additive class

$$f(x) = \sum_{i=1}^{d} \varphi_i(x_i) \tag{13}$$

sits inside the span of a single KAN layer, while an MLP must approximate it with mixtures. This predicts that the KAN advantage is controlled not by the task name but by *how much of the target is additive*, and that it should fall away as cross-coordinate interaction grows.

10.2. Design, and the control that makes it a test

We build a target with a tunable interaction share λ,

$$y = \sqrt{1-\lambda}\, f_{\text{add}}(x) + \sqrt{\lambda}\, f_{\text{int}}(x) + \varepsilon, \qquad x \sim \mathcal{U}[-2,2]^d, \tag{14}$$

with $d = 8$, additive noise of standard deviation 0.05, and both components normalised to unit variance on a large reference sample before mixing, so λ moves interaction power without moving signal power.

Normalising variance is not sufficient on its own. If f_{add} and f_{int} were drawn from different smoothness classes, λ would also move the intrinsic difficulty of the target, and any crossover could be read off the wrong axis. We therefore take both components from the *same* class, degree-two polynomials: $f_{\text{add}} = \sum_i a_i x_i^2$ is purely additive and $f_{\text{int}} = \sum_{(j,k)} c_{jk} x_j x_k$ is purely cross-term, so the only structural difference between the two ends of the sweep is additivity itself. One consequence is worth stating because it is easy to mistake for evidence: the target lies in the span of the degree-two feature set $\{1, x_i, x_i^2, x_j x_k\}$ at *every* λ, so an ordinary least-squares fit on that set is correctly specified throughout, its bias is zero, and its error varies only through estimation variance at fixed feature count. Measured, it is flat to 0.138 dB at $n = 256$ and 0.176 dB at $n = 4096$, but that flatness is a property of the construction rather than a test of it: the same reference moves by 147.36 dB as soon as the target is given a component outside its span. A reference that cannot fail is not a control.

We therefore use a control that can. We fit a structure-agnostic learner at the *same* 2300-parameter budget as the networks, namely ridge regression on 2300 random Fourier features $\cos(w^\top x + b)$ with w drawn from a fixed Gaussian: every feature mixes all coordinates, so nothing in it prefers additive targets, and its capacity is limited in the same way the networks' is. Across the whole sweep its NMSE moves by 0.474 dB at $n = 256$ and 1.17 dB at $n = 4096$. So there is a residual difficulty gradient, and we report it rather than claim it away: it is 8.6% and 20.5% of the architecture effect measured below at the two sample sizes. It also runs in the direction that works against the reading we draw: the interaction end is mildly *easier* for a model that ignores structure, yet it is the end where the KAN loses.

Table 8: Additivity experiment. Paired NMSE difference (dB) between each KAN instantiation and a matched MLP, 10 seeds; negative means the KAN is better. λ is the share of signal power carried by cross-coordinate interaction, with target difficulty held fixed (see text). Bold marks intervals excluding zero in favour of the MLP.

n	λ				
	0.00	0.25	0.50	0.75	1.00
$n = 256$ *training samples*					
KAN (spline)	−3.55	−1.80	−0.33	**+0.93**	**+1.97**
KAN (Fourier)	−12.61	−3.60	−0.52	**+1.53**	**+3.24**
KAN (RBF)	−9.97	−3.51	−0.65	**+1.25**	**+2.20**
KAN (Chebyshev)	−9.31	−3.31	−0.61	**+1.26**	**+2.77**
$n = 4096$ *training samples*					
KAN (spline)	−10.18	−6.61	−5.30	−4.46	−4.75
KAN (Fourier)	−8.28	−4.65	−3.27	−2.38	−2.28
KAN (RBF)	−8.10	−6.23	−5.80	−5.52	−6.62
KAN (Chebyshev)	−6.44	−3.99	−3.23	−2.88	−3.30

All models are sized to the same budget of 2300 parameters, trained with the same schedule, and evaluated on 4000 held-out points over 10 seeds; we report paired differences with bootstrap confidence intervals, so each cell compares models trained on identical data.

10.3. Result

Table 8 reports 40 paired comparisons, so we state the multiplicity treatment before reading it: the intervals are per-cell and uncorrected, and we mark a cell only when its interval excludes zero. The cells the argument rests on, $\lambda \geq 0.75$ at $n = 256$, have magnitudes of 1.97 to 3.24 dB against interval half-widths below half a decibel, so they survive a Bonferroni correction over all 40 cells; the near-zero cells at $\lambda = 0.5$ would not, and we do not build on them. Table 8 shows the predicted ordering, and shows it for all 4 instantiations. At $n = 256$ the KAN advantage decays monotonically in λ and *changes sign*; on purely interacting targets the matched MLP wins by 1.97 to 3.24 dB, with every interval excluding zero. Where exactly the sign flips depends on how the comparison is matched, and we locate it in Section 10.4 rather than reading it off this table alone. Spline, Fourier, RBF and Chebyshev edges differ in how large the advantage is at $\lambda = 0$ but not in where it vanishes, which is what a claim about function class rather than about a particular basis should look like.

At $n = 4096$ the same erosion is present, from 10.18 dB at $\lambda = 0$ to 4.75 dB at $\lambda = 1$ for the spline-KAN, but it saturates instead of crossing zero. The regime map therefore has *two* axes, not one: interaction sets the *size* of the edge-wise advantage, and the training budget decides whether that advantage ever reverses.

Table 9: Tuning control at $n = 4096$. Paired NMSE difference (dB) between the spline-KAN and the *best* swept MLP, with both models given the same schedule; negative means the KAN is better.

Schedule	$\lambda = 0.50$	$\lambda = 1.00$
60 epochs	−5.30	−3.83
180 epochs	−5.52	−3.21

Because a KAN that wins at every λ invites the obvious objection, we ran the control that the objection deserves. Table 9 sweeps MLP depth at $n = 4096$, takes the *best* MLP, and gives both models the same longer schedule. Tripling the schedule closes only 0.62 dB of the gap at $\lambda = 1$ (from 3.83 to 3.21 dB), so the large-sample result is not an artefact of an under-tuned baseline.

10.4. Matching depth, and where the sign actually flips

Equations (2)–(4) size models to a common parameter budget, but budget is not the only thing a comparison has to match. The MLP used above places two hidden layers between input and output, while the KANs place one; the parameter counts agree and the depths do not. That matters here because our own sweep (Table 5) measures depth on the regression task at a scale comparable to several of the effects we are discussing, and because the mismatch runs in the direction that flatters the architecture under test.

We therefore repeat the additivity sweep at two matched depths: a one-hidden-layer MLP against the two-layer KAN, and the two-hidden-layer MLP against a three-layer KAN, both sized to the same 2300-parameter budget. The qualitative result is unchanged at both depths. At $n = 256$ the paired difference moves from −2.86 dB at $\lambda = 0$ to +4.39 dB at $\lambda = 1$ with one hidden layer, and from −1.86 to +2.17 dB with two; at $n = 4096$ the KAN wins at every λ in both cases (−4.43 and −6.13 dB at $\lambda = 1$).

The quantitative correction is to the location of the crossover. Once depth is matched, the MLP is already ahead at $\lambda = 0.5$, by +1.1 dB at one hidden layer and +0.39 dB at two, whereas the depth-mismatched comparison of Table 8 still favours the KAN slightly there. The sign change is therefore at or below $\lambda = 0.5$, not above it, and the unmatched comparison was placing it too late. We report the matched numbers as the ones to use.

10.5. Tuning the other side, where the other side loses

Section 10 already gives the MLP a depth sweep and a longer schedule at $n = 4096$, which is where the KAN wins. The symmetric obligation is to tune the KAN at $n = 256$, which is where it loses, and the first version of this work did not discharge it.

We sweep 8 KAN configurations at the matched budget, four Fourier basis orders and four spline grid sizes, at $\lambda = 0.75$ and $\lambda = 1$. The best configuration in both cases is `spline G=10` , and it does not close the gap: the matched MLP still wins by 0.81 dB at $\lambda = 0.75$ and 1.7 dB at $\lambda = 1$, with both intervals excluding zero. Tuning the KAN buys roughly as much as tuning the MLP bought at the other end of the data axis, and in neither case does it change who wins.

10.6. Reading the wireless regimes through it

The experiment is synthetic by design: it buys a controlled axis at the cost of realism, and it speaks about function class, not about radio physics. Read against it, the three regimes fall where the hypothesis would put them. Memoryless amplitude distortion of the Saleh form is close to additive once the signal is expressed in amplitude and phase, which is the regime where the Fourier-KAN is competitive at matched capacity. Modulation classification from raw I/Q samples is the opposite case: the discriminative information lives in relations *between* samples, not in any single coordinate, and that is where flat KANs lose to both a matched MLP and a small convolutional network. We stress the direction of this argument: the additivity experiment explains the ordering we measured, it does not license a prediction for an unseen task without first estimating how additive that task is.

11. Discussion

11.1. A regime map

Combining the regimes, the audit, and the ablations yields a compact map with three tiers. *Where KANs help:* at higher SNR in modulation classification, over the full SNR range, in the low-data regime, and, with a richer basis, on smooth regression at matched capacity, a KAN is the better flat model, sometimes by a clear margin. Their learnable edges are a real tool, not a gimmick. *Where they do not:* at the default budget and basis, at low SNR, and with plentiful data, a same-size MLP is as good or better, so a KAN advantage is never automatic. *Where architecture matters more than either:* on structured I/Q classification a small CNN with the right inductive bias beats every flat model by a wide margin, so if the task has exploitable structure, the budget is better spent on the prior than on the edge parameterization.

That recommendation owes a word on the axis we have just insisted on. Section 9 argues that inference cost, not parameter count, is the axis that matters for deployment, and on that axis the CNN is the most expensive model in this study, not the cheapest: it takes 25.157 ms per batch, 153.4× the matched MLP and 7.25× the slowest KAN, while holding far fewer parameters. We recommend it anyway, and the ground is accuracy at an inference cost that is still small in absolute terms for a task whose symbol rate is orders of magnitude slower than 25.157 ms per batch of 256. What we do not do is recommend it on the parameter count, which is the metric this paper argues against; a reader whose deployment is latency-bound rather than accuracy-bound should read Table 7 before taking this advice. Layered on top, KAN interpretability is real but not free: the edges are genuinely simple, yet the symbolic extraction that would make them interpretable costs more than a decibel.

11.2. Interpreting the crossovers

The crossovers are not random, and two mechanisms are consistent with all our observations, though we frame them as hypotheses rather than claims. First, the KAN's per-edge nonlinearity is most useful when the target relationship is a clean low-dimensional function of each input. At high SNR the received constellation is sharp, so the mapping from I/Q samples to class is close to a smooth geometric partition that per-edge functions can carve; at low SNR the same flexibility fits noise, which is exactly where the more constrained MLP, and far more so the translation-equivariant CNN, generalize better. This matches both the SNR crossover and the low-data crossover: fewer, cleaner examples favor the model whose inductive bias is a smooth univariate decomposition. Second, on the smooth regression task the benefit of the Fourier basis grows with its order K rather than with width, because a power-amplifier AM/AM curve is well represented by a few harmonics; this is why K=8 overtakes the MLP while wider low-K KANs do not. Neither mechanism supports a blanket advantage; both predict a bounded regime, which is what we measure.

11.3. Why the literature reports sweeping wins

Our results both confirm that KANs can win and explain why the reported wins are overstated. A KAN with an unmatched budget, or with a richer basis than the MLP it is compared to, will beat that MLP, and our own ablations reproduce exactly such a win once the KAN is tuned; attributing it to "the architecture" rather than to the extra effective capacity is the error. Likewise, a saturated benchmark makes every model tie so any fluctuation reads as a win, and an aggregate over a favorable band hides the crossover we measured. The advantage does not vanish under matched control, but it shrinks to a specific, honest set of regimes rather than the blanket superiority usually claimed.

11.4. Threats to validity

We tested three wireless regimes with two KAN variants and one dataset per regime, and the controlled additivity experiment of Section 10 with 4 variants; a broader sweep on the wireless tasks themselves (more datasets, Wav-KAN [12], larger budgets) would refine the regime boundaries, and we release the harness to make that cheap.

Four limits of the controlled experiment deserve to be stated plainly rather than left to be discovered. First, its difficulty control is imperfect: the structure-agnostic reference of Section 10 moves by 1.17 dB across the sweep at $n = 4096$, which is 20.5% of the effect we attribute to additivity there, so at the large sample size a fifth of the measured decay may be difficulty rather than structure. Second, the synthetic family is narrow: one function class (degree-two polynomials) at one input dimension ($d = 8$), chosen so that additivity is the only structural variable, at the cost of saying nothing about smoothness, sparsity or heavy tails. Third, the main tables match parameters but not depth; Section 10.4 shows the ordering survives matching and that the crossover moves when it does, but the wireless tables themselves are not depth-matched. Fourth, seed counts are not uniform across the paper: the regression regimes use 10 seeds against 10 elsewhere and 20 for the paired re-test of Section 8, and we would not quote a sub-decibel difference from a 10-seed cell without the paired interval that accompanies it. The DPD PA is simulated, though from a standard non-polynomial model, so the classical baseline is not privileged; a measured PA would strengthen the regression result. The AMC study is capped in sample count for tractability. Because our conclusions are two-sided, the main risk is mislocating a boundary rather than a hidden global winner: every reported crossover has intervals that exclude zero, and the interpretability cost was measured directly rather than inferred.

11.5. Future work

Three extensions would sharpen the map. First, a *systematic audit* of the recent KAN-in-wireless literature against the six-point checklist of Section 11: our results predict that the reported efficiency wins will correlate with unmatched capacity, saturated benchmarks, or unreported seeds, and a structured survey scoring each paper on the six criteria would test that prediction directly and quantify how widespread the confounds are. Second, a full DPD linearization loop with ACPR and EVM on a *measured* power amplifier, replacing our forward Saleh model, would confirm whether the thin regression crossover survives real memory and hysteresis effects. Third, extending the model set to Wav-KAN [12] on the wireless tasks themselves and adding a real multimodal benchmark such as DeepSense 6G [36] would probe whether any KAN variant reverses the ordering on a task with genuinely exploitable structure. The released harness is designed to make each of these a small addition rather than a reimplementation.

11.6. Practical guidance

For a wireless practitioner: if the task has exploitable structure, spend the model budget on the right inductive bias (a convolution, an unrolled algorithm) rather than on the edge parameterization, which dominated every flat model here. Among flat models, reach for a KAN when labeled data is scarce or when the operating regime is clean (high SNR), where it beat the matched MLP, and tune its basis richness, not just its width; otherwise a same-size MLP is a strong and simpler default. If interpretability is the goal, measure the accuracy cost of the symbolic extraction rather than assuming it is free. The recurring discipline is to match capacity and resolve the operating point before believing any architecture comparison.

11.7. A reporting checklist

The confounds we removed are cheap to check, and we distill them into a checklist a reviewer or author can apply to any KAN-versus-baseline wireless claim:

1. **Matched capacity.** Are the KAN and its baseline sized to the same parameter count, reported explicitly? An efficiency claim against a smaller-parameter baseline is not evidence about the architecture.
2. **Non-saturated benchmark.** Is the reported operating point one where models are below the ceiling? If every model exceeds $\sim$99%, the comparison is uninformative.

3. **Resolved operating point.** Is accuracy resolved by SNR (or the relevant axis) rather than aggregated over a band that may favor one model, as our crossover shows?
4. **Multiple seeds with intervals.** Is the claim supported by a paired difference with a confidence interval that excludes zero, not a single-seed number?
5. **Strong baseline.** Is the baseline the one a practitioner would use (a CNN for I/Q, a memory polynomial for DPD), not a deliberately weak MLP?
6. **Interpretability with fidelity.** If interpretability is claimed, is the accuracy cost of the symbolic extraction measured, not just a plot of a few edges?

Every "win" we could not reproduce under matched control failed at least one of these checks; every genuine KAN win we found survives all six.

12. Reproducibility

Every number in this paper is produced by a script and checked against a machine-readable claims file. The harness (`harness.py`) implements the KAN variants, the MLP, the matched-capacity sizing, the training loop, and the bootstrap statistics; `exp_dpd.py`, `exp_amc.py`, and `exp_interp.py` produce the main tables and Figs. 2–7, and `abl_dpd.py` and `abl_amc.py` produce the ablations (Table 6, Figs. 8–10), each from its own result CSV. The experiments added in this version follow the same convention: `exp_mechanism2.py` and `exp_mechanism2_tuning.py` produce Tables 8 and 9 of Section 10, `exp_amc_backbone.py` produces the shared-backbone comparison of Section 5.4, `exp_mlp_sweep_all.py` the extended symmetric sweep of Section 8.1, and `exp_bigbudget_paired.py` the paired recomputation reported in Section 8; `variants.py` holds the FastKAN and Cheby-KAN layers. The RML2016.10a dataset is public [28], and `prep_rml_band.py` documents the exact SNR band and the content hash of the array used.

The layout of the released material is flat and fixed: experiment scripts and the harness in `code/`, every result CSV in `code/results/` with one file per experiment, raw run logs in `code/results/logs/`, the manuscript and its figures in `paper/`, and one number per file in `paper/claims/` with `paper/claims.json` as the single-source-of-truth index. Two tables of Section 10 are themselves generated from the CSVs by `code/sinh_bang_coche.py`, so re-running an experiment changes the printed table rather than leaving it stale, and `code/verify_numbers.py` checks the manuscript text against `claims.json`, including that any superlative in the prose points at the extremal row of the group it names. We fix all random seeds. Everything above is available at `https://github.com/haodpsut/kan-wireless-benchmark`.

13. Conclusion

We asked when Kolmogorov–Arnold networks help in wireless, and answered by measurement. Under a parameter-matched, multi-seed, non-saturated protocol, the honest answer is two-sided: KANs are competitive and win in specific regimes, at higher SNR in modulation classification, in the low-data regime, and with a richer basis on smooth regression, but they never dominate, are beaten decisively by the right inductive bias on structured tasks, lose to a same-size MLP at the default budget and at low SNR, and carry an interpretability cost of more than a decibel to extract. The sweeping "efficiency wins" reported in the wireless KAN literature are consistent with unmatched capacity and saturated or favorably chosen benchmarks rather than with a blanket architectural advantage. We hope the regime map, the protocol, and the released harness make the next such claim easier to place, and to check, before it is believed.

CRediT authorship contribution statement

Phuc Hao Do: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to assist with code review, refactoring of the experimental harness, and copy-editing of the manuscript. The author reviewed and edited all output and takes full responsibility for the content of this article. No generative tool was used to produce, alter, or interpret any experimental measurement reported here.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The code, the data-preparation scripts, and the result tables that reproduce every number reported in this article are openly available at `https://github.com/haodpsut/kan-wireless-benchmark` (see Section 12 for the mapping from each table and figure to the script and result file that produce it).

References

[1] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, M. Tegmark, KAN: Kolmogorov-Arnold Networks, arXiv preprint arXiv:2404.19756 (2024).

[2] Blealtan, efficient-kan: An efficient pure-pytorch implementation of kolmogorov-arnold networks, `https://github.com/Blealtan/efficient-kan` (2024).

[3] B. Khalid, F. Cavaliere, L. Giorgi, P. Freire, S. K. Turitsyn, J. E. Prilepsky, DPD-KAN: Kolmogorov-Arnold Networks for Low Complexity Digital Predistortion in 5G Analog Radio-over-Fiber Systems, arXiv preprint arXiv:2606.21178 (2026).

[4] H. Chen, R. Zhou, Q. Yuan, Z. Guo, W. Fu, KAN-ResNet-enhanced radio frequency fingerprint identification with zero-forcing equalization, Sensors 25 (7) (2025) 2222.

[5] H. Zhang, F. Zhou, H. Du, Q. Wu, C. Yuen, Revolution of wireless signal recognition for 6g: Recent advances, challenges and future directions, arXiv preprint arXiv:2503.08091 (2025).

[6] R. Yu, W. Yu, X. Wang, KAN or MLP: A Fairer Comparison, arXiv preprint arXiv:2407.16674 (2024).

[7] S. Somvanshi, S. A. Javed, M. M. Islam, D. Pandit, S. Das, A survey on kolmogorov-arnold network, arXiv preprint arXiv:2411.06078 (2024).

[8] A. N. Kolmogorov, On the representation of continuous functions of several variables by superpositions of continuous functions of one variable and addition, Doklady Akademii Nauk SSSR 114 (1957) 953–956.

[9] G. Cybenko, Approximation by superpositions of a sigmoidal function, Mathematics of Control, Signals and Systems 2 (4) (1989) 303–314.

[10] K. Hornik, Approximation capabilities of multilayer feedforward networks, Neural Networks 4 (2) (1991) 251–257.

[11] Z. Liu, P. Ma, Y. Wang, W. Matusik, M. Tegmark, KAN 2.0: Kolmogorov-Arnold Networks Meet Science, arXiv preprint arXiv:2408.10205 (2024).

[12] Z. Bozorgasl, H. Chen, Wav-KAN: Wavelet Kolmogorov-Arnold Networks, arXiv preprint arXiv:2405.12832 (2024).

[13] Z. Li, FastKAN: Very Fast Implementation of Kolmogorov-Arnold Networks, arXiv preprint arXiv:2405.06721 (2024).

[14] S. SS, K. AR, G. R, A. KP, Chebyshev polynomial-based kolmogorov-arnold networks, arXiv preprint arXiv:2405.07200 (2024).

[15] B. Khalid, P. Freire, S. K. Turitsyn, J. E. Prilepsky, Hardware-oriented inference complexity of kolmogorov-arnold networks, arXiv preprint arXiv:2604.03345 (2026).

[16] A. Paszke, S. Gross, F. Massa, et al., PyTorch: An imperative style, high-performance deep learning library, in: Advances in Neural Information Processing Systems (NeurIPS), 2019.

[17] C. Rudin, Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead, Nature Machine Intelligence 1 (5) (2019) 206–215.

[18] C. Molnar, Interpretable Machine Learning, Independently published, 2020.

[19] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems (NeurIPS), 2017.

[20] G. Hinton, O. Vinyals, J. Dean, Distilling the knowledge in a neural network, arXiv preprint arXiv:1503.02531 (2015).

[21] J. Ba, R. Caruana, Do deep nets really need to be deep?, in: Advances in Neural Information Processing Systems (NeurIPS), 2014.

[22] T. J. O'Shea, J. Hoydis, An introduction to deep learning for the physical layer, IEEE Transactions on Cognitive Communications and Networking 3 (4) (2017) 563–575.

[23] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, Nature 521 (7553) (2015) 436–444.

[24] C. D. Fontes da Silva, T. Jiang, L. Zhang, V. Bobrovs, X. Yu, X. Pang, O. Ozolins, E. Porto da Silva, Low complexity kolmogorov-arnold network-based digital predistortion for analog radio-over-fiber fronthaul, arXiv preprint arXiv:2606.27042 (2026).

[25] J. Chen, Y. Lv, W. Zhao, L. Xin, G. Xu, T. Liu, Real-valued time-delay kolmogorov-arnold network for digital predistortion of rf power amplifiers, in: International Conference on Microwave and Millimeter Wave Technology (ICMMT), 2025.

[26] S. Chen, Y. Zu, Z. Feng, S. Yang, M. Li, Y. Ma, J. Liu, Q. Pan, X. Zhang, C. Sun, RadioLLM: Introducing large language model into cognitive radio via hybrid prompt and token reprogrammings, arXiv preprint arXiv:2501.17888 (2025).

[27] T. J. O'Shea, J. Corgan, T. C. Clancy, Convolutional radio modulation recognition networks, in: Int. Conf. on Engineering Applications of Neural Networks (EANN), 2016, pp. 213–226.

[28] T. J. O'Shea, N. West, Radio machine learning dataset generation with gnu radio, in: Proceedings of the GNU Radio Conference, 2016.

[29] T. J. O'Shea, T. Roy, T. C. Clancy, Over-the-air deep learning based radio signal classification, IEEE Journal of Selected Topics in Signal Processing 12 (1) (2018) 168–179.

[30] A. Krizhevsky, I. Sutskever, G. E. Hinton, Imagenet classification with deep convolutional neural networks, in: Advances in Neural Information Processing Systems (NeurIPS), 2012.

[31] V. Sathyanarayanan, P. Gerstoft, A. El Gamal, RML22: Realistic dataset generation for wireless modulation classification, IEEE Transactions on Wireless Communications 22 (11) (2023) 7663–7675.

[32] H. Zhang, F. Zhou, Q. Wu, C. Yuen, FSOS-AMC: Few-shot open-set learning for automatic modulation classification over multipath fading channels, arXiv preprint arXiv:2410.10265 (2024).

[33] L. Ding, G. T. Zhou, D. R. Morgan, Z. Ma, J. S. Kenney, J. Kim, C. R. Giardina, A robust digital baseband predistorter constructed using memory polynomials, IEEE Transactions on Communications 52 (1) (2004) 159–165.

[34] D. R. Morgan, Z. Ma, J. Kim, M. G. Zierdt, J. Pastalan, A generalized memory polynomial model for digital predistortion of rf power amplifiers, IEEE Transactions on Signal Processing 54 (10) (2006) 3852–3860.

[35] A. A. M. Saleh, Frequency-independent and frequency-dependent nonlinear models of twt amplifiers, IEEE Transactions on Communications 29 (11) (1981) 1715–1720.

[36] A. Alkhateeb, G. Charan, T. Osman, A. Hredzak, J. Morais, U. Demirhan, N. Srinivas, DeepSense 6G: A large-scale real-world multi-modal sensing and communication dataset, IEEE Communications Magazine 61 (9) (2023) 122–128.

[37] A. K. Gizzini, Y. Medjahdi, A. J. Ghandour, L. Clavier, Towards explainable ai for channel estimation in wireless communications, arXiv preprint arXiv:2307.00952 (2023).

[38] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: International Conference on Learning Representations (ICLR), 2015.